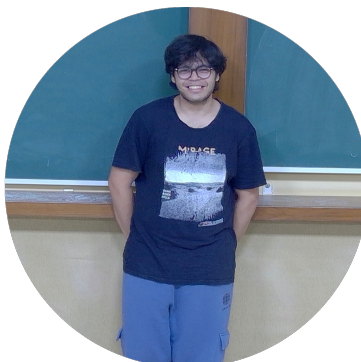




Dhairya Baxi

Postgraduate Student • HiCAPS Lab (Prof. Uday Bondhugula)
Department of Computer Science and Automation
Indian Institute of Science, Bangalore

About

I am an M.Tech student in Computer Science and Automation at IISc, currently working at the **HiCAPS Lab** under Prof. Uday Bondhugula. My primary interest lies in **AI compilers**. I enjoy working on compiler optimization and polyhedral compilation for modern hardware. I'm also broadly interested in mathematics and theoretical computer science.

Outside of coursework and research, I have co-founded the [Recreational Mathematics Club \(RMC\)](#) at IISc.

Education

Indian Institute of Science, Bangalore

Master of Technology · Computer Science and Engineering (2025 – Present)
Department of Computer Science and Automation

Institute of Technology, Nirma University

Bachelor of Technology · Computer Science and Engineering (2021 – 2025)

Experience

Polymage Labs

AI Compiler Intern · May 2026 – July 2026

Bengaluru, Karnataka, India

- Implemented a register tiling heuristic for CPUs, improving attention workload performance by up to 26.8% (14.1% geomean) across varying batch sizes, head counts, and sequence lengths.
- Designed and implemented a new NVIDIA CUDA Tile backend, including a conversion pass from affine loops to CUDA Tile IR.

Skills: MLIR, C++, GPUs, Performance Analysis

Scalable Compilers for Heterogeneous Architectures Group, IIT Hyderabad

Research Intern · December 2024 – August 2025

Hyderabad, Telangana, India

- Worked on compiler optimization projects under the guidance of Dr. Ramakrishna Upadrasta.
- Developed, refactored, and benchmarked MLIR passes and related infrastructure.
- Performed performance evaluation, plotting, and automation scripting.

Skills: C++, MLIR, torch-mlir, ISL, Barvinok

TuskerAI

Summer Research Intern · April 2024 – July 2024

Ahmedabad, Gujarat, India

- Developed **Quantum-Gated GRU (QG-GRU)**, integrating quantum circuits with traditional GRUs.
- Conducted experiments and prepared results for presentation.

Skills: Qiskit, PyTorch, Python, Matplotlib

Publication

PolyUFC: Polyhedral Compilation Meets Roofline Analysis for Uncore Frequency Capping

Nilesh Rajendra Shah, M V V S Manoj Kumar, *Dhairya Baxi*, and Ramakrishna Upadrasta

2026 IEEE/ACM International Symposium on Code Generation and Optimization (CGO)

[Google Scholar](#) · [PDF](#)

Projects

Polyqsim · [GitHub](#)

QASM2 to MLIR Affine for statevector quantum circuit simulation

Polyqsim is a tool that translates QASM2 quantum assembly code into the MLIR Affine dialect, enabling integration of quantum programs into MLIR polyhedral optimization infrastructure for efficient statevector simulation.

FianchettoAD · [GitHub](#)

Automatic Differentiation

FianchettoAD is a programming language that supports automatic differentiation (autodiff) out of the box. With FianchettoAD, you can easily compute derivatives of functions without having to implement differentiation manually. The language is currently under development and is incomplete.

Math visualizers in HTML5 canvas

- [Computing pi with Collisions](#)
- [Twin Dragon Curve](#)
- [Cardioid Visualization](#)

Contact

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Google Scholar	scholar.google.com/citations?user=TM4x-mwAAAAJ

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